Nanyang Technological University
School of Electrical & Electronic Engineering
IE2106 Engineering Mathematics I

Tutorial 4 Probability & Statistics

1. (a) A chemical supply company currently has in stocks 100 kg of a certain chemical, which it sells to customers in 5-kg lots. Let X = the number of lots ordered by a randomly chosen customer, and suppose that X has probability mass function given by

x	1	2	3	4
$P(x)$	0.2	0.4	0.3	0.1

Compute $E(X)$ and $Var(X)$. Hence, compute the expected number of kilograms left after the next customer's order is shipped, and the variance of the number of kilograms left.

(b) Let X be a random variable with probability density function given by

$$f(x) = \begin{cases} cx^2, & 0 \leq x \leq 1 \\ 0, & otherwise \end{cases}$$

where c is a constant. Find c, then compute the mean and variance of X and X^2. Hence, write down the values of $E(2X+1)$, $Var(2X+1)$, $E(3-2X^2)$ and $Var(3-2X^2)$.

2. Timeless is a company that manufactures quartz crystal watches. Timeless researchers have shown that the watches have an average life of 28 months before certain electronic components deteriorate, causing the watch to become unreliable. The standard deviation of watch lifetimes is 5 months, and the distribution of lifetimes is normal.

(a) If Timeless guarantees a full refund on any defective watch for 2 years after purchase, what percentage of total production will they expect to give refund?

(b) If Timeless does not want to make refunds on more than 12% of the watches they make, how long should the guarantee period be (to the nearest month)?

3. The number of people who arrive at a cashier's counter in POSB during a specified period of time possesses (approximately) a Poisson probability distribution. Suppose the mean number of arrivals per minute for cashier service at POSB is one person per minute. What is the probability that in a given minute, the number of arrivals will equal three or more? Can you tell the bank manager that the number of arrivals will rarely exceed three per minute?

Answers

1.(a) 2.3, 0.81, 88.5, 20.25
(b) $c = 3$, $\frac{3}{4}$, $\frac{3}{80}$, $\frac{3}{5}$, $\frac{12}{175}$, $\frac{5}{2}$, $\frac{3}{20}$, $\frac{9}{5}$, $\frac{48}{175}$

2.(a) 21.19% (b) 22 months

3. 0.0803